THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 8 — MATHS · NCERT-TYPE UNIT TEST SET 1

Chapters 6–10: **We Distribute, Yet Things Multiply, Proportional Reasoning-1, Fractions in Disguise, The Baudhayana–Pythagoras Theorem, Proportional Reasoning-2** | RBSE Pattern | Session 2026–27

Time: 2 Hours Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–15 are Objective Type (1 mark each).
3. Q. 16–20 are Very Short Answer Type (2 marks each).
4. Q. 21–25 are Short Answer Type (3 marks each).
5. Q. 26–27 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 24–25 and Q. 26–27.
7. Write the question number before each answer.
8. NCERT-type: questions in the style of the textbook examples and exercises.

SECTION — A (Objective Type Questions) 15 × 1 = 15

Choose the correct option and write it in your answer book.

1. $(x+3)^2$ equals: 1

(a) x^2+3x+9 **(b)** x^2+6x **(c)** x^2+6x+9 **(d)** x^2+9

2. The ratio of 2 m 40 cm to 60 cm is: 1

(a) 4 : 2 **(b)** 1 : 4 **(c)** 4 : 1 **(d)** 5 : 1

3. A value changes from 80 to 100. The percentage increase is: 1

(a) 22% **(b)** 50% **(c)** 28% **(d)** 25%

4. The legs of a right triangle are 5 cm and 7 cm. Its hypotenuse (in cm) is: 1

(a) 12 **(b)** $\sqrt{74}$ **(c)** $5\sqrt{3}$ **(d)** $2\sqrt{3}$

5. If A:B = 2:3 and B:C = 4:5, then A:B:C is: 1

(a) 8:12:15 **(b)** 2:7:5 **(c)** 2:3:5 **(d)** 8:15:8

6. $(x-2)^2$ equals: 1

(a) x^2-2x+4 **(b)** x^2+4x+4 **(c)** x^2-4 **(d)** x^2-4x+4

7. If A : B = 2 : 3 and B : C = 6 : 5, then A : B : C is: 1

(a) 2 : 6 : 5 **(b)** 4 : 6 : 5 **(c)** 2 : 3 : 5 **(d)** 2 : 3 : 30

8. 18 is what percent of 72? 1

(a) 25% **(b)** 22% **(c)** 50% **(d)** 28%

9. The legs of a right triangle are 7 cm and 9 cm. The area of the square drawn on its hypotenuse (in cm²) is: 1

(a) 16 **(b)** 63 **(c)** 130 **(d)** 256

10. 18 men can do a piece of work in 10 days. 12 men will do it in (days): 1

(a) 15 **(b)** 180 **(c)** 18 **(d)** 11

11. $(x+3)(x-3)$ equals: 1

(a) x^2-6x+9 **(b)** x^2+6x-9 **(c)** x^2+9 **(d)** x^2-9

12. On a map, 1 cm represents 5 km. A distance of 3.5 cm on the map is actually: 1

(a) 1.75 km **(b)** 17.5 km **(c)** 22.5 km **(d)** 8.5 km

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13. A cycle bought for ₹ 1500 is sold for ₹ 1800. The gain percent is: 1

(a) 18% **(b)** 20% **(c)** 40% **(d)** 22%

14. Taking $m = 8$ in $(2m,\ m^2 - 1,\ m^2 + 1)$, the Pythagorean triplet is: 1

(a) 16, 63, 66 **(b)** 8, 63, 65 **(c)** 16, 64, 65 **(d)** 16, 63, 65

15. On a map the scale is 1 cm = 15 km. Two places 135 km apart are shown on the map at a distance (in cm) of: 1

(a) 2025 **(b)** 10 **(c)** 9 **(d)** 120

SECTION — B (Very Short Answer Type) — 5 × 2 = 10

Answer in about 20–30 words each.

16. Expand $(x + 8)^2$ using an identity. 2

17. Find the ratio of 3 m 60 cm to 90 cm in simplest form. 2

18. What percent of 90 is 36? 2

19. Find the Pythagorean triplet with the smallest member 28. 2

20. Which ratio is greater: 5:9 or 3:5? 2

SECTION — C (Short Answer Type) — 5 × 3 = 15

Answer in about 40–60 words each.

21. Expand $(2x + 3)^2$. 3

22. If A : B = 2 : 3 and B : C = 6 : 5, find A : B : C. 3

23. The marked price of a laptop is ₹ 45000. GST at 18% is added. Find the GST amount and the final bill. 3

24. The hypotenuse of an isosceles right triangle is $4\sqrt{2}$ cm. Find the length of each equal side and the area of the triangle. 3

OR

On a map of scale 1 cm = 2 km, a rectangular park measures 7 cm by 4 cm. Find the actual length, breadth and perimeter of the park. 3

25. Find 53×47 using the identity $(a + b)(a - b) = a^2 - b^2$. 3

OR

10 men can build a wall in 18 days. How many days will 15 men take to build it? 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

26. A rectangle has length $(x + 3)$ cm and breadth $(x + 5)$ cm. Write an expression for its area and find the area when $x = 4$. 5

OR

Divide ₹ 1200 among A, B and C in the ratio 1 : 3 : 4. If C gives ₹ 100 to A, find the new ratio of their amounts. 5

27. Find the compound interest on ₹ 8000 for 3 years at 10% per annum compounded annually. By how much does it exceed the simple interest for the same period? 5

OR

The diagonals of a rhombus are 40 cm and 42 cm. Find its side, perimeter and area. 5

END OF QUESTION PAPER **Total: 50 Marks**

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